Calderón–Zygmund Decomposition

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Overview

1. Motivation

2. The Calderón–Zygmund decomposition

3. An application: continuity of the Hilbert Transform

Motivation: Truncating functions

Let $f \in L^1(\mathbb{R}^n)$, and pick a level $\alpha > 0$.

$$S^- := \{ x : |f(x)| \leq \alpha \},$$
$$S^+ := \{ x : |f(x)| > \alpha \}.$$

- **On S^-:** $|f(x)| \leq \alpha$, so $f(x)$ is automatically bounded.
- **On S^+:** $f(x)$ may be large, but Chebyshev's inequality gives

$$\mu(S^+) \;=\; \mu(\{ x : |f(x)| > \alpha \}) \;\leq\; \frac{1}{\alpha} \int_{\mathbb{R}^n} |f(x)|\, dx.$$

Hence, the measure of S^+ is controlled by $\|f\|_1/\alpha$, indicating S^+ is "small".

This naive technique is still often useful, and is a starting point for more refined decompositions.

The Calderón–Zygmund Decomposition Theorem

Theorem (For $f \in L^1(\mathbb{R}^n)$ and $\alpha > 0$)

There exist $\Omega = \bigcup Q_k$ union of semi-disjoint cubes, and a decomposition $f = g + b$, with $b = \sum_k b_k$ such that:

1. $\|g\|_{L^\infty} \leq 2^n \alpha$
2. $\|g\|_1 \leq \|f\|_1$
3. supp $b_k \subseteq Q_k$
4. $\int_{Q_k} b_k = 0$
5. $\|b_k\|_{L^1} \leq 2^{n+1} \alpha \mu(Q_k)$
6. $\mu(\Omega) \leq \alpha^{-1} \|f\|_1$

The Idea —— Choosing the cubes (I)

Goal: Choose cubes in such a way that the L^1 norm on each cube is controlled by its measure

Easy: Recursively bisect cubes, cherry-picking those that work

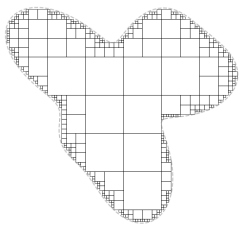
Tricky: Prove what is left outside the cubes is a.e. bounded

Formal Construction: Choosing Cubes

- Find a diameter large enough such that for $\left\{Q_k^{(0)}\right\}$, the lattice of parallel cubes of identical size,

$$\frac{1}{\mu(Q_k^{(0)})}\int_{Q_k^{(0)}} |f| \leq \alpha,$$

which exists because $f \in L^1$.



- We now create new cubes by dyadic subdivision. Consider a cube $Q_k^{(m)}$: if

$$\frac{1}{\mu(Q_k^{(m)})}\int_{Q_k^{(m)}} |f| > \alpha,$$

we keep this cube, otherwise we skip it.

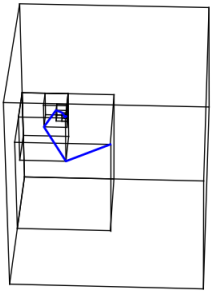
Let $\Omega := \bigcup_k Q_k$, where $\{Q_k\}$ are all those chosen in this process.

Formal Construction: Splitting Cubes (II)

Question: what happens to points that are never chosen?

Consider a Lebesgue point (of f) $x \in \mathbb{R}^n$ that is in no cube chosen in this process, that is $x \in \Omega^c$.

Then
there exists a sequence $(Q_{k_m}^{(m)})$ such that $x \in Q_{k_m}^{(m)}$ for all m.



Since $Q_{k_m}^{(m)}$ is a sequence that *shrinks nicely*
to x, by the Lebesgue differentiation theorem, we must have

$$|f(x)| = \lim_{m\to\infty} \frac{1}{\mu(Q_{k_m}^{(m)})} \int_{Q_{k_m}^{(m)}} |f| \leq \alpha$$

Formal Construction: Calculations (III)

- Suppose $Q_k^{(m+1)}$ is cut out of $Q_{k'}^{(m)}$, we then have

$$\alpha < \frac{1}{\mu(Q_k^{(m+1)})} \int_{Q_k^{(m+1)}} |f| \leq \frac{1}{2^{-n}\mu(Q_{k'}^{(m)})} \int_{Q_{k'}^{(m)}} |f| \leq 2^n \alpha,$$

  because $Q_{k'}^{(m)}$ was skipped.

Formal Construction: Good and Bad functions (IV)

- Let

$$b_k = \chi_{Q_k}\left(f - \frac{1}{\mu(Q_k)}\int_{Q_k} f\right)$$

  and $b = \sum_k b_k$.
- Define then

$$g := f - b$$

All desired propertied follow trivially from the construction, as

$$g(x) = \begin{cases} f(x) & x \in \Omega^c \\ \frac{1}{\mu(Q_k)}\int_{Q_k} f & x \in Q_k \end{cases}$$

Application to Singular Integrals: the Hilbert Transform

Definition (Hilbert Transform)

The transform $H : \mathcal{S}(\mathbb{R}) \to \mathcal{S}(\mathbb{R})$ is defined by its Fourier multiplier as follows:

$$\widehat{Hf}(\xi) = -i \cdot \operatorname{sgn}(\xi) \cdot \widehat{f}(\xi)$$

This operator is used extensively in the theory of Fourier Multipliers, and serves as a perfect illustration of the principle at play.

Theorem (Hilbert transform continuity)

The Hilbert transform is bounded from L^p to itself, for $p > 1$.

Definition (Weak L^p)

Let λ be the distribution operator, then L^p_* is the space of functions f such that:

$$\|f\|_{L^p_*} := \left[\sup_{\alpha} \alpha^p \lambda(f)(\alpha)\right]^{\frac{1}{p}} < \infty$$

Definition (Weak type boundedness)

An operator T is of weak type (p, q) if it is bounded from L^p to L^q_*.

Definition (Strong type boundedness)

An operator T is of strong type (p, q) if it is bounded from L^p to L^q.

Hilbert Transform: a preliminary reduction

Proposition

If Hilbert transform is of weak type $(1,1)$*, then it is of strong type* (p,p)*.*

Proof.

Suppose H is weak $(1,1)$. By Plancherel's Theorem, H is strong (and thus weak) $(2,2)$, thus by Marcinkiewicz's interpolation theorem it is of strong type (p,p) for all $1 < p < 2$. Since H is self-adjoint, for $1 < p' < 2$:

$$\|Hf\|_p = \sup_{\|g\|_{p'} \leq 1} |\langle Hf, g\rangle| = \sup_{\|g\|_{p'} \leq 1} |\langle f, Hg\rangle| \leq \|f\|_p \sup_{\|g\|_{p'} \leq 1} \|Hg\|'_p \lesssim \|f\|_p$$

□

Proof of weak $(1,1)(I)$

We only need to show H is of weak type $(1,1)$. Let $f \in \mathcal{S}(\mathbb{R}) \subseteq L^1(\mathbb{R})$.
Fix $\alpha > 0$, and perform the Caldéron-Zygmund decomposition of $\mathbb{R}$, obtaining a collection of dyadic intervals $\{I_k\}$, and a decomposition $f = g + b$.
Let $\lambda(h)$ be the distribution function of the function h, then:

$$\begin{aligned}
\lambda(Hf)(\alpha) &\leq \lambda(Hg)(\alpha/2) + \lambda(Hb)(\alpha/2) \\
&\leq (\frac{\alpha}{2})^{-2} \|Hg\|_{L^2}^2 + \mu(\Omega^*) + \lambda(Hb|_{(\Omega^*)^c})(\alpha/2) \\
&\leq \frac{4}{\alpha^2} \|g\|_{L^2}^2 + \frac{2}{\alpha} \|f\|_{L^1} + \frac{2}{\alpha} \int_{\mathbb{R} \setminus \Omega^*} \sum_k |Hb_k|
\end{aligned}$$

We can now apply different strategies for g and b.

Proof of weak $(1,1)$ (II)

Good part:
We use the fact that g is bounded pointwise a.e. by $2^n\alpha$ and in norm by f:

$$\frac{4}{\alpha^2}\|g\|_{L^2}^2 = \frac{4}{\alpha^2}\int_{\mathbb{R}} |g|^2 \leq \frac{2^{n+2}}{\alpha}\int_{\mathbb{R}} |g| \leq \frac{2^{n+2}}{\alpha}\|f\|_{L^1}$$

Proof of weak $(1,1)$ (II)

Bad part:

We have, for $x \notin 2I_k$

$$Hb_j(x) = \frac{1}{\pi} \int_{I_k} \frac{b_k(y)}{x-y} dy$$

And, remembering b_k has null mean,

$$\begin{aligned}
\int_{\mathbb{R}\setminus 2I_k} |Hb_k| &= \frac{1}{\pi} \int_{\mathbb{R}\setminus 2I_k} \left| \int_{I_k} \frac{b_k(y)}{x-y} dy \right| dx \\
&= \frac{1}{\pi} \int_{\mathbb{R}\setminus 2I_k} \left| \int_{I_k} b_k(y) \left(\frac{1}{x-y} - \frac{1}{x-c_k} \right) dy \right| dx \\
&\leq \frac{1}{\pi} \int_{I_k} |b_k(y)| \left(\int_{\mathbb{R}\setminus 2I_k} \left| \frac{1}{x-y} - \frac{1}{x-c_k} \right| dx \right) dy \\
&\leq \frac{1}{\pi} \int_{I_k} |b_k(y)| \left(\int_{\mathbb{R}\setminus 2I_k} \frac{\mu(I_k)}{|x-c_k|^2} dx \right) dy \leq \frac{2}{\pi} \|b_k\|_{L^1}
\end{aligned}$$

Conclusion

We can now assert

$$\begin{aligned}
\lambda(Hf)(\alpha) &\leq \frac{4}{\alpha^2}\|g\|_{L^2}^2 + \frac{2}{\alpha}\|f\|_{L^1} + \frac{2}{\alpha}\int_{\mathbb{R}\setminus\Omega^*}\sum_k |Hb_k| \\
&\leq \frac{2^{n+2}}{\alpha}\|f\|_{L^1} + \frac{2}{\alpha}\|f\|_{L^1} + \frac{2}{\alpha}\sum_k \|Hb_k\|_{L^1(\mathbb{R}\setminus 2I_k)} \\
&\leq \frac{2^{n+2}}{\alpha}\|f\|_{L^1} + \frac{2}{\alpha}\|f\|_{L^1} + \frac{4}{\pi\alpha}\sum_k \|b_k\|_{L^1} \\
&\leq \frac{2^{n+2}}{\alpha}\|f\|_{L^1} + \frac{2}{\alpha}\|f\|_{L^1} + \frac{16}{\pi\alpha}\|f\|_{L^1} \\
&\lesssim \frac{1}{\alpha}\|f\|_{L^1},
\end{aligned}$$

thus proving our result. □

Thank you

Thank you!